Unit: HIT339 Distributed Development

Project: TopEndLibraryHub
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1 My Responsibility

My duties included implementing branches, reception desks, and interbranch transfers. Additionally, I developed instructions for implementation and tried to test implementation using Oracle VirtualBox. Purpose: To help staff locate items and move them between library branches.

2 Branches and Reception Desks

The application implements three fictional branches.

Branch	Branch code	Reception desk
Darwin Hub	DAR	DAR-D01
Palmerston Hub	PAL	PAL-D01
Casuarina Hub	CAS	CAS-D01

Every reception desk belongs to one of the branches. Books, music, and toys are in a single item-location relationship which defines their physical branch.

Existing demonstration stock has been allocated to Darwin Hub during migration. The branch and desk codes are unique database indexes. The seed method checks existing codes before insertion.

3 Staff Working Location

Admin and Reception users select the working branch and reception desk from the Working location page.

The application ensures that the desk belongs to the selected branch and they are both active. The selected location is attached to the staff session and displayed under the navigation bar.

New inventory items get the Admin's validated working branch. Existing inventory edits leave item's location unchanged. Staff uses the transfer workflow to move stock between branches.

4 Transfer Workflow

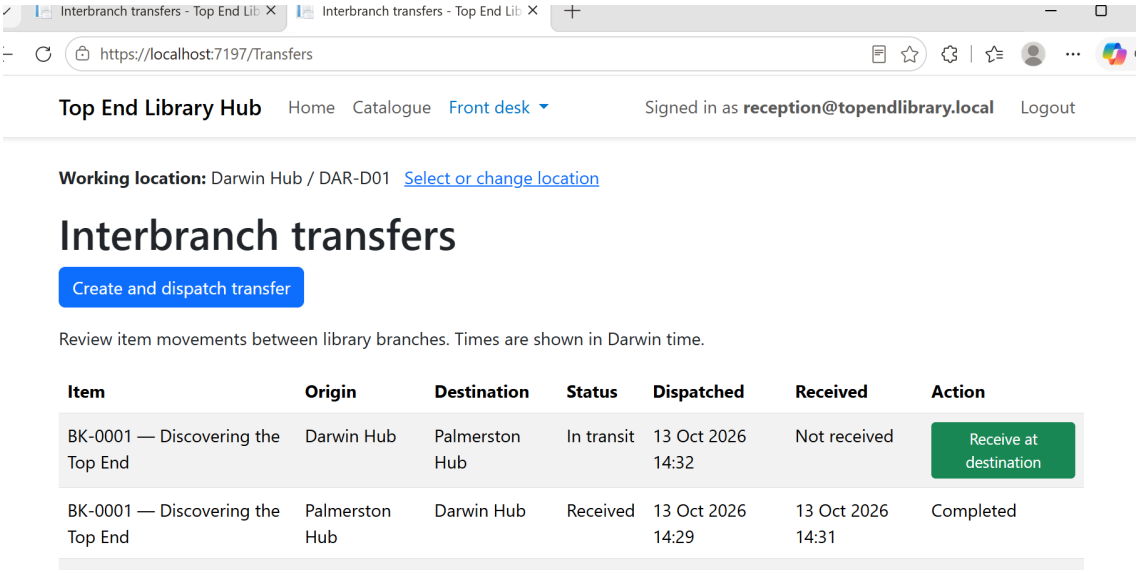
Staff selects an eligible item at the current branch and another active destination. Dispatch creates a transfer record and makes the item In transit.

The transfer registers item, origin, destination, desks, staff members, dispatch and receipt timestamps. During transit, the item maintains its origin branch and the active transfer registers its destination. Checkout is unavailable during transit. Only staff at the destination branch can receive the item. The receipt operation updates item's location and finishes the transfer. The transfer list displays its travel history and timestamps in Darwin time zone.

5 Functional Testing Evidence

The following screenshots were taken on 13 October 2026. They illustrate the previous implementation and not the integrated holds one.

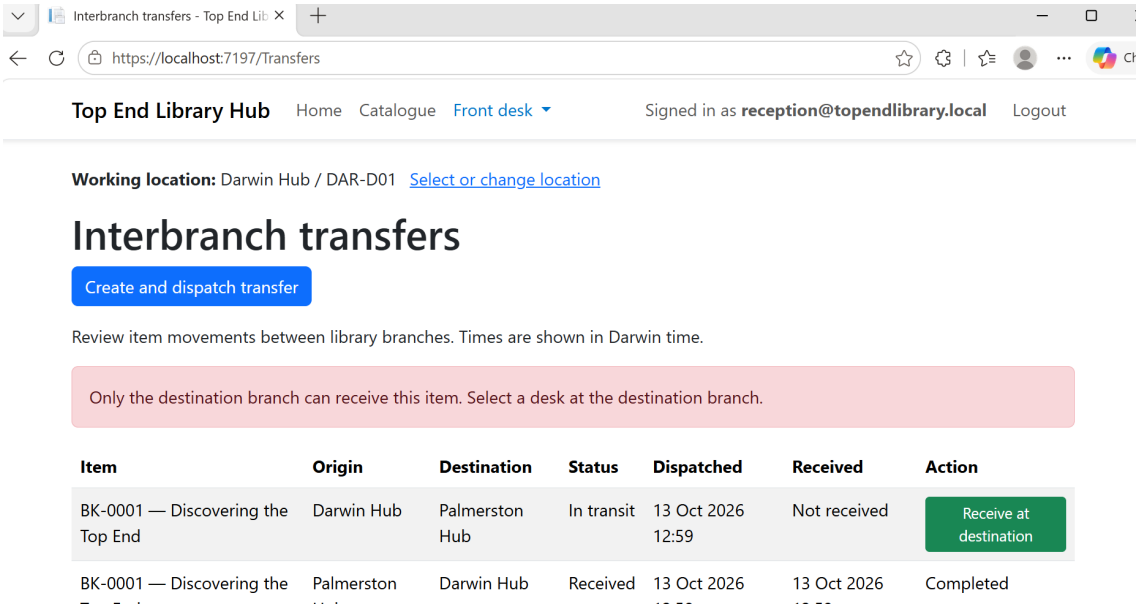
5.1 Dispatching an Item



BK-0001 dispatched from Darwin Hub to Palmerston Hub.

The transfer list showed the item, origin, destination, dispatch timestamp and In transit status.

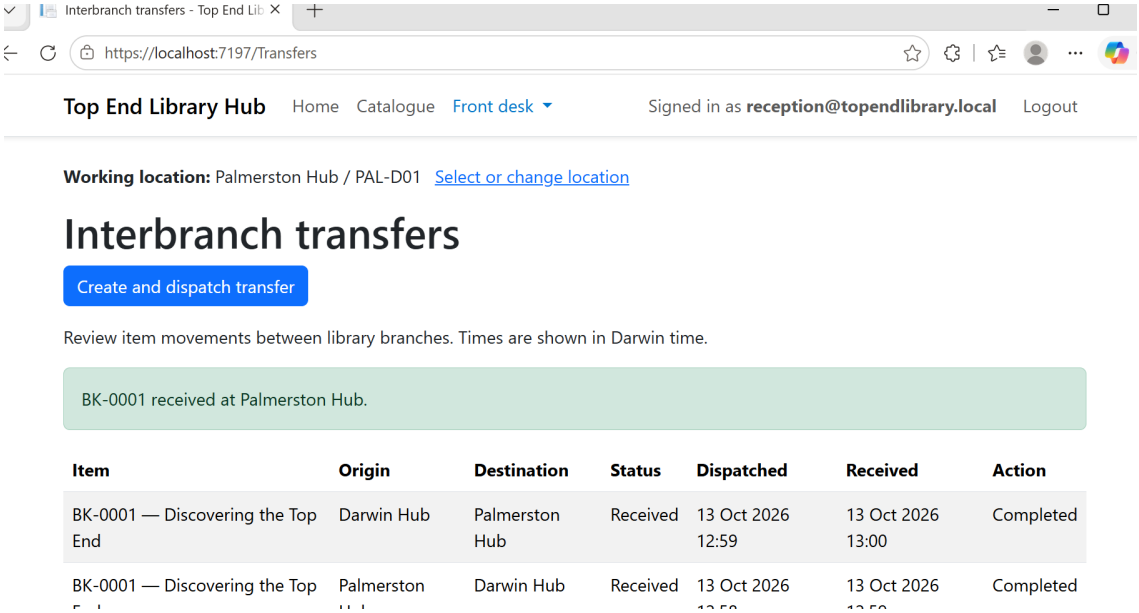
5.2 Rejecting Receipt at the Wrong Branch



Receipt refused while staff works at the origin branch.

The application explained that the destination branch is required for receipt. The transfer was still in transit.

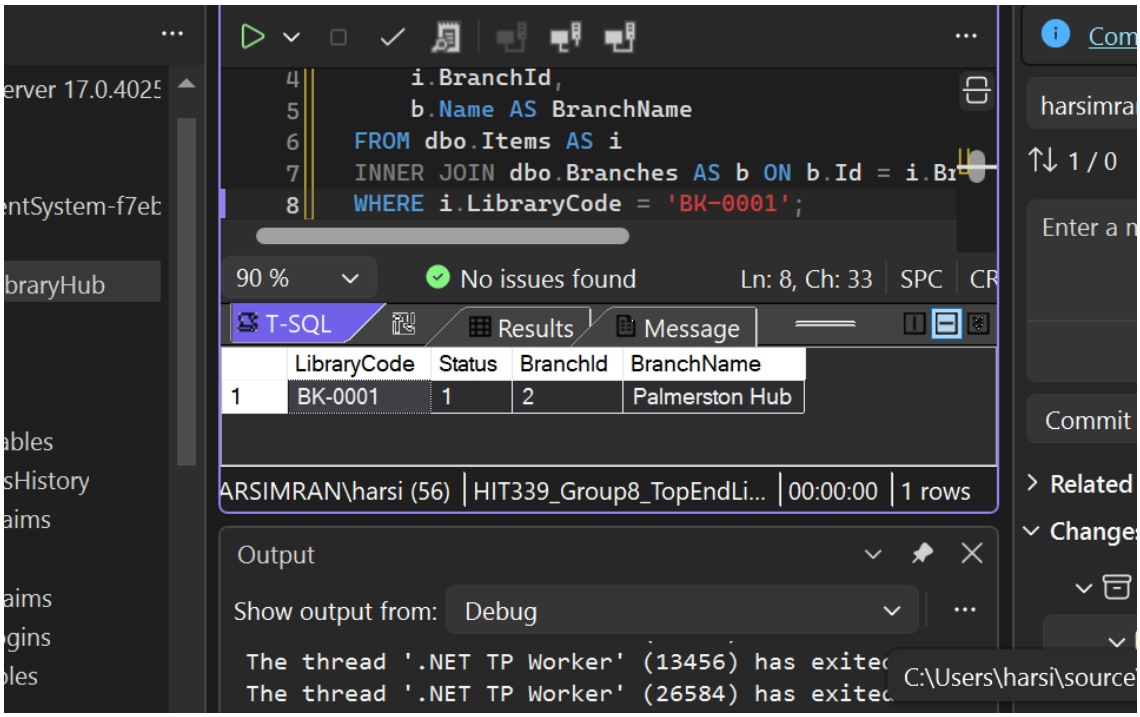
5.3 Receiving at the Destination Branch



Receipt completed at Palmerston Hub.

The application displayed a success message, Received status, receipt timestamp and Completed action.

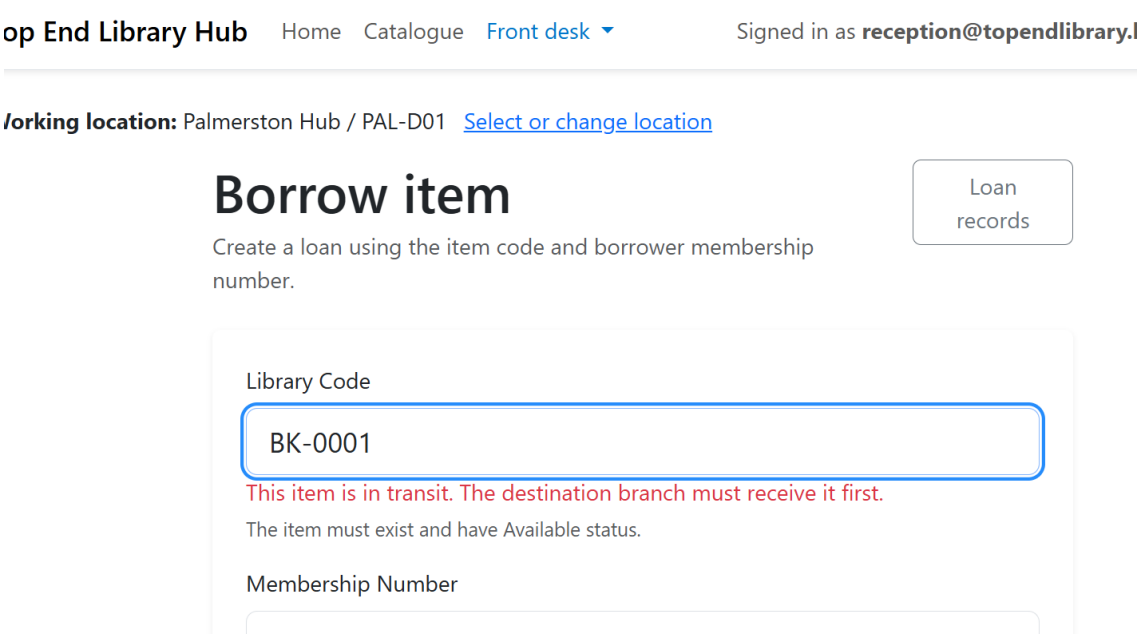
5.4 Confirming the Updated Item Location



Confirmation from the database about the item location update.

The database query result showed BK-0001 Available at Palmerston Hub with branch ID = 2.

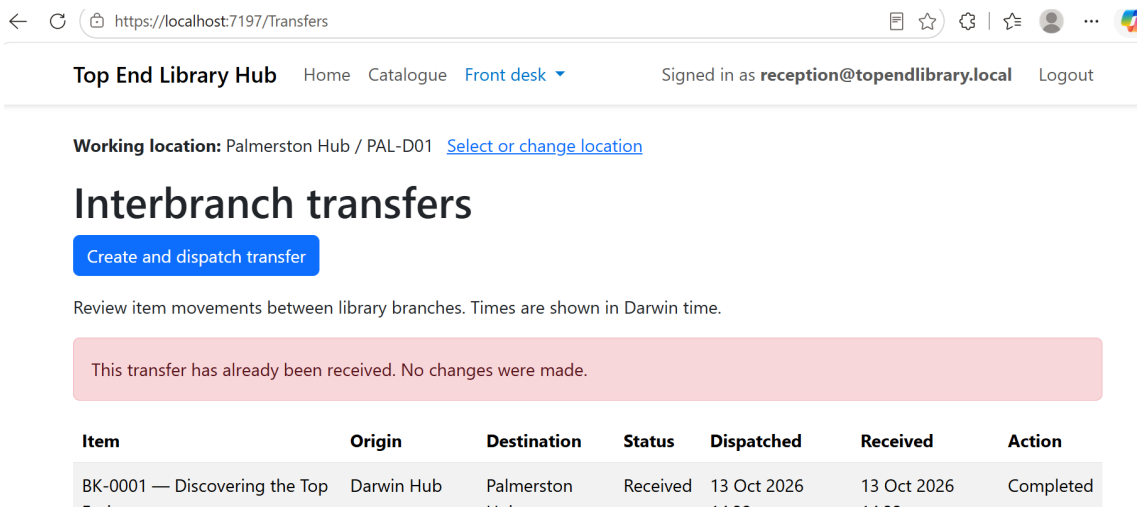
5.5 Preventing Checkout During Transit



Checkout refused while the item is In transit.

The borrowing form displayed a message requiring the destination receipt first.

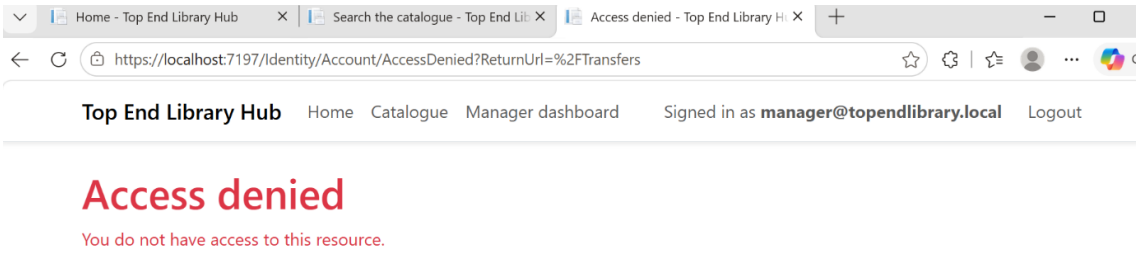
5.6 Rejecting Repeat Receipt



Receipt Request Rejected as Repeat

The system showed an error that the transfer had already been received, and there were no changes made. It needs to be independently verified if the timestamp for the receipt hasn't changed using the database.

5.7 Restricting Manager Access to Transfers



Manager account denied access to transfers.

The Manager account received an Access denied page while opening the transfer list.

5.8 Test Summary

Test	Observed result
Dispatch an eligible item	Transfer displayed as In transit
Receive at the wrong branch	Rejected
Receive at the destination branch	Completed successfully
Update item location	Palmerston location confirmed Borrow during transitRejected
Repeat receipt	Already received message displayed
Manager accesses transfers	Access denied

Invalid branch/desk combinations, repeat seeding, dispatch to the same branch, duplicate dispatch, inventory creation/edit location preservation still require recording actual results.

6 Holds and Notification Integration

Kamalpreet's integrated implementation associates each hold with the pickup branch.

Code review shows that dispatch operation checks the allocation and pickup destination. Receipt operation refreshes the pick-up-readiness of the hold inside transaction and requests a notification after the transaction is committed.

Checkout and transfers use shared item locking. Checkout also checks the item location, transit state and hold ownership.

The integrated branch compiles and runs successfully on my laptop. Runtime log shows database queries and simulated notification activity.

The end-to-end held-item transfer test and checkout/dispatch simultaneous testing still incomplete. Successful compilation and running does not prove that the workflows have been implemented correctly.

7 Installation and Running Guide

7.1 Prerequisites

The application requires Windows, Git, .NET 10 SDK, SQL Server Express LocalDB and the browser. Visual Studio can be used to open, compile and run the project.

The recorded development versions are:

Component	Version
.NET SDK	10.0.400
ASP.NET Core runtime	10.0.11
Entity Framework Core packages	10.0.10
Target framework	.NET 10

7.2 Installation Process

1. Install the prerequisites.

2. Download the team's agreed repository branch.

3. Record the application commit being tested.

4 . Load the solution or project into Visual Studio.

5. Restore dependencies.

6. Verify database connection settings.

7. Build and launch the application.

8. Wait until the startup initializer applies committed migrations and seeds demonstration data.

9. Open the local add9. Open the local address shown by the application.

10. Sign in and select a valid branch and reception desk.

The final guide must specify the agreed branch, commit, solution filename and any additional configuration for the integrated modules.

7.3 Running at Different Branches

All the locations use the same code and installation process. Staff selects their location from the Working location page.

After selecting the branch and reception desk, the staff can access the transfer list, dispatch eligible stock and receive incoming items at the right destination.

7.4 Restart Verification

Once you have restarted the application, ensure that your old data can be accessed and seeding does not generate duplicates of branches and desks.

8 Virtual Machine Testing

VM installation was tried using Oracle VirtualBox.

Darwin VM installed and verified the .NET SDK and Git installation successfully. However, SQL Server 2022 LocalDB installation failed because SQL Server VSS Writer service could not start. Administrator Installation failed as well. The Event Viewer indicated the start-up timeout.

The Canberra VM installation attempt showed the administrator restriction. Underlying SQLWriter failure was not confirmed.

VM check Status

.NET SDK installation on Darwin Completed

Git installation on Darwin Completed

LocalDB installation Blocked

9 Task and Hours Record

The estimates below are recommended planning estimates. The actual hours are unknown and must be taken from my work record.

Task	Estimated hours
Branch and reception-desk models	2
Database migration and seed data	3
Working-location selection and display	3
Transfer dispatch and receipt	5
Checkout and transfer validation	3
Book, Music and Toy compatibility changes	3
Functional testing and evidences	3
Installation guide and report	3
VM setup and troubleshooting problem	4

Integration and compile review	2
Total	31

The troubleshooting time must be counted in actual hours even if the installation attempt was unsuccessful.

10 Strengths and Weaknesses

The strengths of my contribution include implementing a clear dispatch/receipt workflow and invalid actions check along with the successful transfers. I also ensured my work visibility via feature-branch commits and screenshots.

The limitation with the contributions I have made is that the proper VM was arranged at the last moment, and there were problems with installation, hence making it impossible to validate the guide.

11 Learning

This task gave me an understanding how changing shared model affects existing application features. Creating the required branch relationship also needed inventory creation/edit location preservation. I have learned about the need to keep existing data during migrations, validate staff locations on the server and maintain transfer history.

Code review of the integration gave me an understanding of the reasons for sharing locks and availability rules for checkout, reservations and transfers. I learned how to distinguish implemented behavior, code review and testing supported by actual evidence.

12 Value Added

My contribution allows tracking item movement across branches. The staff can see the item's traveling branch, transfer status and dispatch/receipt timestamp.

The working-location display shows the selected branch and reception desk. The validation messages explain why action is refused such as receipt at the wrong branch or checkout during transit. Inventory compatibility changes allow to assign the valid branch to the new item. Ordinary edits leave item's existing location unchanged.

13 Final Handoff Status

My initial branch and transfer work is kept in harsimran-branches. Kamalpreet's shared branch contains later holds and circulation integration.

The unfinished work includes successful VM installation testing, integrated holds/notifications testing, simultaneous operations testing and outstanding validation evidence.

The guide and test record must be checked against the final agreed application version. Only tests with actual results recorded must be considered as passed.